

CBSE Practice Worksheet

Class 7 | Mathematics | Difficulty: Hard | Set E

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Simplify: $(a^2 - b^2)/(a + b)$
(A) $a - b$
(B) $a + b$
(C) $a^2 - b$
(D) $a - b^2$
2. Find: $(-1)^{1000} + (-1)^{999}$
(A) 0
(B) 2
(C) -2
(D) 1
3. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$
(A) $x - 2$
(B) $x + 2$
(C) $(x-2)^2$
(D) $x^2 - 4$
4. Solve: $2^{x+1} = 32$
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
5. Solve: $\sqrt{x + 7} = x - 5$
(A) $x = 9$
(B) $x = 2$
(C) $x = 9$ or 2
(D) No solution
6. Solve: $|2x - 5| = 9$
(A) $x = 7$ or -2
(B) $x = 7$ or 2
(C) $x = -7$ or 2
(D) $x = -7$ or -2
7. If $2^x = 3^y = 6^z$, which is true?
(A) $1/z = 1/x + 1/y$
(B) $z = x + y$
(C) $z = xy$
(D) $z = x - y$
8. If $a:b = 3:4$ and $b:c = 5:6$, find $a:c$
(A) 3:6
(B) 5:8
(C) 15:24

(D) 1:2

9. Factorize: $x^2 - 5x - 14$

(A) $(x-7)(x+2)$

(B) $(x+7)(x-2)$

(C) $(x-7)(x-2)$

(D) $(x+7)(x+2)$

10. If $\log_{\blacksquare} 8 = x$, find x

(A) 2

(B) 3

(C) 4

(D) 8

Section B: Fill in the Blanks (1 mark each)

11. If $a + b + c = 0$, then $a^3 + b^3 + c^3 = \underline{\hspace{2cm}}$

12. If $x^2 - 6x + 8 = 0$, then $x = \underline{\hspace{2cm}}$ or $\underline{\hspace{2cm}}$

13. $(x + 1/x)^2 - (x - 1/x)^2 = \underline{\hspace{2cm}}$

14. Rationalize: $1/(\sqrt{3} + 1) = (\sqrt{3} - 1)/\underline{\hspace{2cm}}$

15. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - \underline{\hspace{2cm}}$

16. If $5^{\blacksquare} = 125$, then $x = \underline{\hspace{2cm}}$

17. $\text{HCF} \times \text{LCM}$ of 12 and 18 = $\underline{\hspace{2cm}}$

18. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 = \underline{\hspace{2cm}}$

Section C: Short Answer Questions (2 marks each)

19. Solve: $|x - 3| + |x + 2| = 7$

20. If $x = 2 + \sqrt{3}$, find $x + 1/x$

21. Find the value of x : $3^{(2x-1)} = 27^{(x-2)}$

22. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$

23. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

24. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

25. Simplify: $(x^{\blacksquare} - 16)/(x^2 + 4)$

--- End of Worksheet ---

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